

The computer is built by combining chemical components into a solution in a tube. Various small DNA molecules are mixed in the solution with selected DNA enzymes and ATP. The latter is used as the energy source of the device. The molecules start interacting upon one another. By tinkering with the type of DNA and enzymes in the mix, scientists can fine-tune the process to a desired result.

Retinal implant enables blind to see

A team of German scientists led by Eberhart Zrenner has found in a chip a possible cure for retinitis pigmentosa. Implantation of the chip is largely comparable to standard vitreous or retinal surgery, as performed in cases of complicated retinal detachment or lesions. The core of the implant is a microchip (approximately 3 mm in diameter and 50µm thick), with an array of roughly 1500-pixel fields.



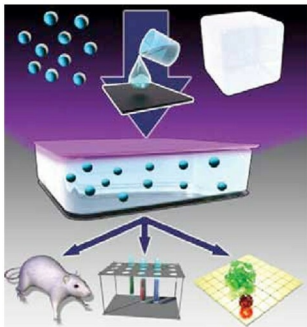
Microchip array

Photocells, an amplifying circuit and a stimulation electrode are attached to each pixel field. The photocells absorb the light entering the eye, transforming it into electrical signals. A tiny power line provides energy from an external source behind the ear. Sixteen additional electrodes are placed for testing purpose.

The chip is capable of taking over the function of light-sensitive cells (photoreceptors) which perish in the degenerative retinal disease. The implant is placed onto that retinal layer where light-sensitive cells are present, thus making use of the remaining natural-information-processing channels of the retina.

Anti-radiation chip

Synergy Environics has introduced Enviro Chip, which changes the



Embedding nanocrystals in glass provides a way to create UV-producing LEDs for biomedical applications

nature of radiation from mobile phones and computers on human beings. The passive device operates at frequencies between 500 and 600 THz.

Operating at a much higher frequency, the chip generates random waves, which become the carrier waves over the constant microwaves emitted from mobile phones, computers and other WiFi devices and make these compatible with the body's random waveforms. Thus it acts as a radiation protector when fixed on the device.

The Enviro Chip varies in size as it is designed for different devices such as mobile phones, laptops and desktop computers. It is fixed on a specific location on the external surface of the device.

Texting collar for cows

As part of a joint project, The University of Strathclyde, Morrisons, Scottish Agricultural College (SAC), NMR, Harbro and Well Cow and Embedded Technology Solutions are developing an intelligent collar device for cows.

The device picks up subtle movements with the same sensor as used in Wii gaming and generates a continuous

record of the cows' activity patterns. Results are sent back using a range of wireless technologies like 3G, with a full update accessible via a hub or even a mobile phone. Farmers can set their phone to receive a text alert when a cow is in distress or entering labour.

By accurately sensing the cow's head positions in three dimensions, the collar is able to detect whether the animal's hind legs have begun to lower—one of the initial signs of illness.

Light-emitting nanocrystal diodes go ultraviolet

A multinational team of scientists has developed a process for creating glass-based, inorganic light-emitting diodes (LEDs) that produce light in the ultraviolet range. The work is a step toward biomedical devices with active components made from nanostructured systems.

The glass-based material can emit light in the ultraviolet spectrum, and be integrated onto silicon chips that are the principal components of current electronic technologies.

The devices are inorganic and combine the chemical inertness and mechanical stability of glass with the property of electric conductivity and electroluminescence (i.e. the ability of a material to emit light in response to the passage of an electric current). As a result, these can be used in harsh environments, such as for immersion into physiologic solutions, or by implantation directly into the body.

The active part of the device consists of tin dioxide nanocrystals covered with a shell of tin monoxide embedded in standard glass. By tuning the shell thickness it is possible to control the electrical response of the whole material.

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